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ng to partition the observations of the n -by- p data matrix
 .dx) containing cluster indices of each observation.
 correspond to variables.

[example](#)

distance measure and the [k-means++ algorithm](#) for

cluster indices with additional options specified by one or

[example](#)

number of times to repeat the clustering using new initial

centroid locations in the k -by- p matrix C .

[example](#)

in-cluster sums of point-to-centroid distances in the k -

[example](#)

ances from each point to every centroid in the n -by- k

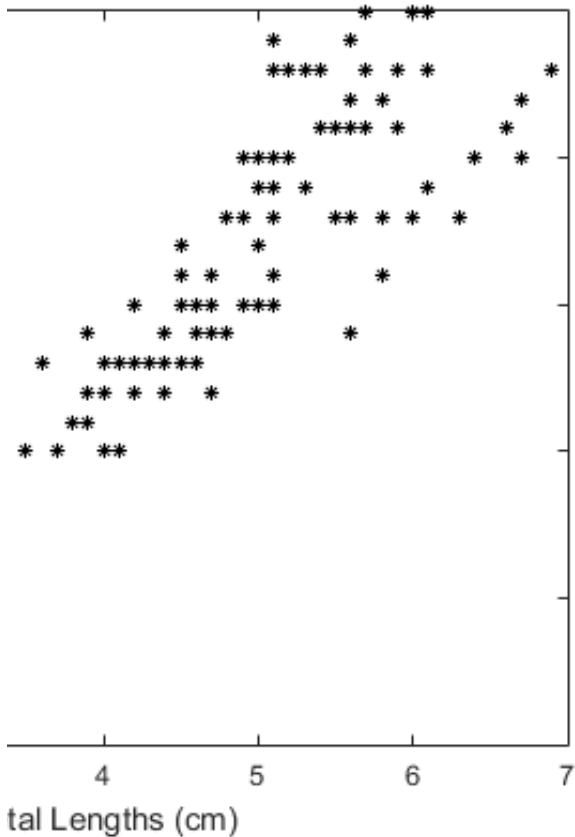
[example](#)[collapse all](#)

lot the cluster regions.

ths and widths as predictors.

```
' ,5);
```

sher's Iris Data



er variance region and a higher variance region. This might indicate
ters.

centroid initialization and squared Euclidean distance by default. It is
a by setting the 'Replicates' name-value pair argument.

responding to the observations in X. C is a 3-by-2 matrix containing

h centroid to points on a grid. To do this, pass the centroids (C) and
e iteration of the algorithm.

a fine grid on the plot

```
h',1,'Start',C);...
```

to the closest centroid

erations.

gorithm did not converge, which you should expect since the software

```

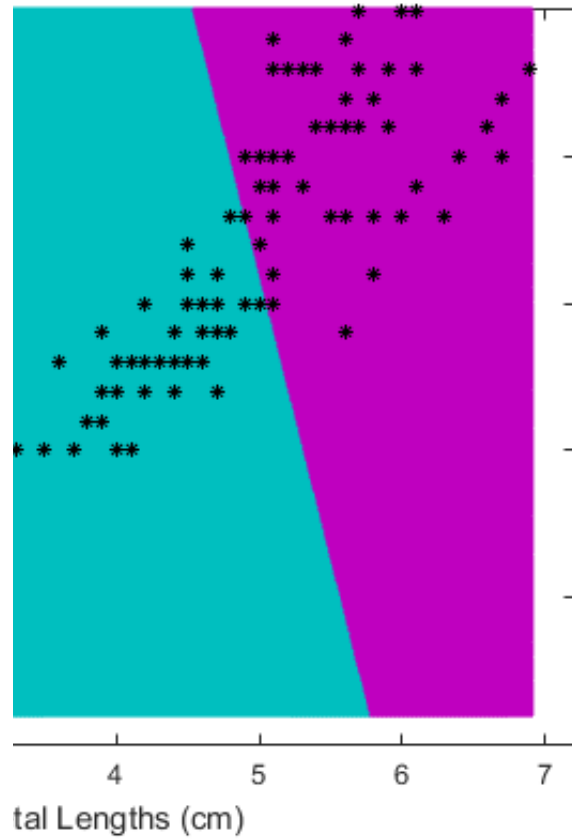
region,...
.75,0],'.');

',5);

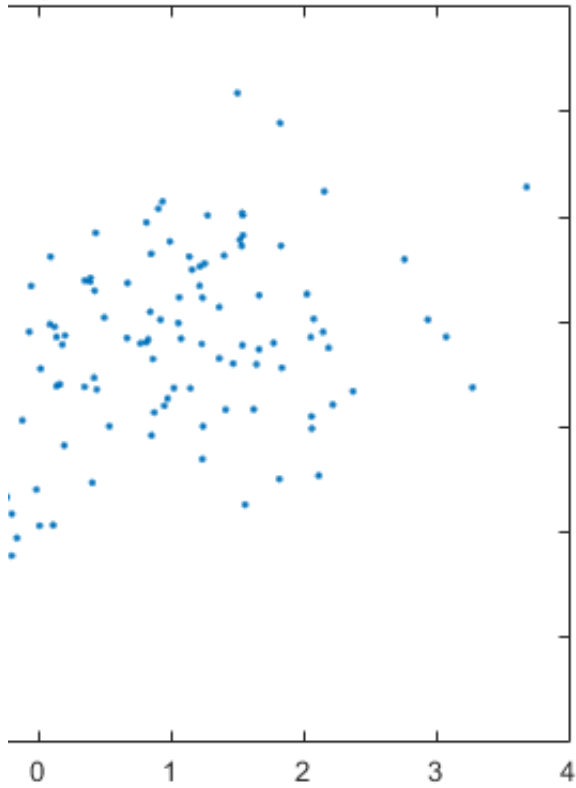
1 3','Data','Location','Best');

```

sher's Iris Data



my Generated Data



the best arrangement out of five initializations. Display the final

```
byblock',...
```

```
1 of distances = 201.533.
1 of distances = 201.533.
1 of distances = 201.533.
1 of distances = 201.533.
1 of distances = 201.533.
}3
```

as separately using *k*-means++.

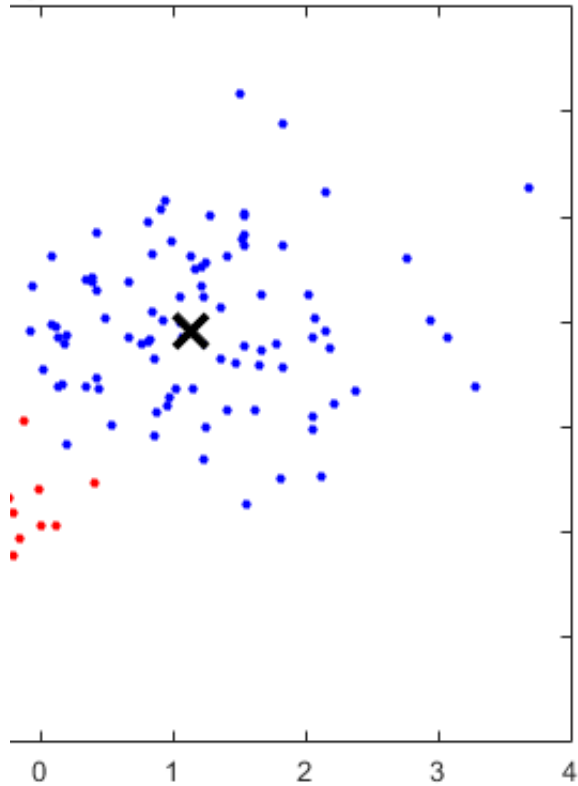
```
MarkerSize',12)
```

```
MarkerSize',12)
```

```
roids',...
```

```
roids'
```

Assignments and Centroids



usters are by passing `idx` to `silhouette`.

rticularly if you use online updates (set by default). If you have a
 u invoke a pool of workers, then `kmeans` runs each clustering task (or
`s` > 1, then the parallel computing decreases time to convergence.

aussian mixture model.

```
3)'); % Gaussian mixture mean
positive-definite covariance
```

del with 20 components. `X` is a 10000-by-30 matrix of data generated

ons for parallel computing.

```
% Invokes workers
% Random number stream
UseSubstreams',1,...
```

ing the 'local' profile ... connected to 2 workers.

stream specifies to use the multiplicative lagged Fibonacci generator
 ing fields that specify options for controlling estimation.

workers are available. The number of workers might vary on your system.

Specify that there are $k = 20$ clusters in the data and increase the
 function contains local minima. Specify 10 replicates to help find a

```
rs',options,'MaxIter',10000,...  
l0);
```

```
sum of distances = 7.58059e+06.  
sum of distances = 7.5904e+06.  
sum of distances = 7.59086e+06.  
sum of distances = 7.57758e+06.  
sum of distances = 7.58614e+06.  
sum of distances = 7.59462e+06.  
sum of distances = 7.57765e+06.  
sum of distances = 7.57639e+06.  
sum of distances = 7.60079e+06.  
sum of distances = 7.58117e+06.  
sum of distances = 7.58117e+06.
```

of iterations and the terminal objective function value for each
 results of replicate 9 because it has the lowest total sum of distances.

[collapse all](#)

of X correspond to observations, and the columns correspond to

is an n -by-1 data matrix, regardless of its orientation.

positive integer.

e, Value arguments. Name is the argument name and Value is the single quotes (' '). You can specify several name and value pair NameN, ValueN.

s',10,'Options',statset('UseParallel',1) specifies the cosine g values, and to use parallel computing.

ndow, specified as the comma-separated pair consisting of %:

ration
in

osine' | 'correlation' | 'hamming'

sed for minimization, specified as the comma-separated pair

for the different, supported distance measures. This table summarizes lae, x is an observation (that is, a row of X) and c is a centroid (a row

	Formula
Euclidean (default). Each element is the mean of the elements in that cluster.	$d(x,c)=(x-c)(x-c)'$
Manhattan (L1). Each element is the sum of the absolute values of the differences between the observation and the centroid.	$d(x,c)=\sum_{j=1}^p x_j-c_j $
Cosine. The distance is the cosine of the angle between the observation and the centroid, treating them as vectors. The mean of the elements in that cluster, ignoring those with zero Euclidean distance, is used to normalize the vectors.	$d(x,c)=1-\frac{xc'}{\sqrt{(xx')(cc')}}$
Dynamic Time Warping. The distance is the minimum sum of absolute differences between the observation and the centroid, allowing for non-linear alignments between the two sequences. The centroid is the mean of the observations.	$d(x,c)=1-\frac{(\vec{x}-\vec{\bar{x}})(\vec{c}-\vec{\bar{c}})'}{\sqrt{(\vec{x}-\vec{\bar{x}})(\vec{x}-\vec{\bar{x}})'}\sqrt{(\vec{c}-\vec{\bar{c}})(\vec{c}-\vec{\bar{c}})'}}$

cent-wise mean in that cluster, g and those points to nd unit iation.	where $\vec{x} = \frac{1}{p} \left(\sum_{j=1}^p x_j \right) \vec{1}_p$ $\vec{c} = \frac{1}{p} \left(\sum_{j=1}^p c_j \right) \vec{1}_p$ $\vec{1}_p$ is a row vector of p ones.
is only inary data. ortion of bits ch centroid is nt-wise ints in that	$d(x, y) = \frac{1}{p} \sum_{j=1}^p I\{x_j \neq y_j\},$ where I is the indicator function.

loses all member observations

observations, specified as the comma-separated pair consisting of
varizes the available options.

	Description
	Treat an empty cluster as an error.
	Remove any clusters that become empty. kmeans sets the corresponding return values in C and D to NaN.
	Create a new cluster consisting of the one point furthest from its centroid (default).

ns

re comma-separated pair consisting of 'MaxIter' and a positive

eparated pair consisting of 'OnlinePhase' and 'on' or 'off'.

s performs an online update phase in addition to a batch update
ing for large data sets, but guarantees a solution that is a local
rds, the software finds a partition of the data in which moving any
total sum of distances.

ive algorithm for minimizing fitting criteria

set

or minimizing the fitting criteria, specified as the comma-separated
array returned by [statset](#). These options require Parallel Computing

	Description
	A RandStream object or cell array of such objects. If you do not specify Streams, kmeans uses the default stream or streams. If you specify Streams, use a single object except when: • You have an open parallel pool • UseParallel is true. • UseSubstreams is false. In that case, use a cell array the same size as the parallel pool. If a parallel pool is not open, then Streams must supply a single random number stream.
	• If true, Replicates > 1, and if a parallel pool of workers from the Parallel Computing Toolbox is open, then the software implements <i>k</i>-means on each replicate in parallel. • If the Parallel Computing Toolbox is not installed, or a parallel pool of workers is not open, computation occurs in serial mode. Default is default, meaning serial computation.
	Set to true to compute in parallel in a reproducible fashion. Default is false. To compute reproducibly, set Streams to a type allowing substreams: 'mlfg6331_64' or 'mrg32k3a'.

[ool](#) and explicitly create a parallel pool before invoking kmeans and
,1).
=1',1)

at clustering using new initial cluster centroid positions

w initial cluster centroid positions, specified as the comma-separated
er. kmeans returns the solution with the lowest [sumd](#).

ying a 3-D array as the value for the 'Start' name-value pair

ster centroid positions

uniform' | numeric matrix | numeric array

sitions (or *seeds*), specified as the comma-separated pair consisting a numeric array. This table summarizes the available options for

	Description
	Perform a preliminary clustering phase on a random 10% subsample of X. This preliminary phase is itself initialized using 'sample'.
	Select k seeds by implementing the k-means++ algorithm for cluster center initialization.
	Select k observations from X at random.
	Select k points uniformly at random from the range of X. Not valid with the Hamming distance.
	k-by-p matrix of centroid starting locations. The rows of Start correspond to seeds. The software infers k from the first dimension of Start, so you can pass in [] for k.
	k-by-p-r array of centroid starting locations. The rows of each page correspond to seeds. The third dimension invokes replication of the clustering routine. Page j contains the set of seeds for replicate j. The software infers the number of replicates (specified by the 'Replicates' name-value pair argument) from the size of the third dimension.

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Close

ta, and removes any row of X containing at least one NaN. Removing

[collapse all](#)

n vector. idx has as many rows as X, and each row indicates the rrvation.

eric matrix. C is a [k-by-p](#) matrix, where row j is the centroid of cluster j.

centroid distances

ances, returned as a numeric column vector. `sumd` is a `k`-by-1 vector, of distances within cluster j .

centroid

eturned as a numeric matrix. `D` is an n -by- k matrix, where element m is the distance from observation m to centroid m .

[collapse all](#)

is an iterative, data-partitioning algorithm that assigns n observations to k clusters, where k is chosen before the algorithm starts.

. For example, choose k observations at random (by using `kmeans` or `kmeanspp` for cluster center initialization (the default).

ances of all observations to each centroid.

l by `OnlinePhase`):

ation to the cluster with the closest centroid.

observations to a different centroid if the reassignment decreases the total within-cluster point-to-cluster-centroid distances.

ances in each cluster to obtain k new centroid locations.

signments do not change, or the maximum number of iterations is reached.

to find centroid seeds for k -means clustering. According to Arthur and Vassilvopoulos, the time complexity of Lloyd's algorithm, and the quality of the final solution.

follows, assuming the number of clusters is k .

from the data set, X . The chosen observation is the first centroid, c_1 .

in to c_1 . Denote the distance observation m is from c_1 as $d(x_m, c_1)$.

in X with probability

observation to each centroid, and assign each observation to its closest

centroid j at random from X with probability

s closest to centroid c_p and x_m belongs to C_p .

er with a probability proportional to the distance from itself to the
 $\geq$.

n.

a simulation study for several cluster orientations, that k -means++
 of within-cluster, sum-of-squares point-to-cluster-centroid distances

to minimize the sum of point-to-centroid distances, summed over

where each iteration consists of reassigning points to their nearest
 by recalculation of cluster centroids. This phase occasionally does not
 nimum. That is, a partition of the data where moving any single point
 al sum of distances. This is more likely for small data sets. The batch
 roximates a solution as a starting point for the second phase.

tes, where points are individually reassigned if doing so reduces the
 ls are recomputed after each reassignment. Each iteration during this
 l the points. This phase converges to a local minimum, although there
 r total sum of distances. In general, finding the global minimum is
 rting points, but using several replicates with random starting points
 global minimum.

he default), then the software selects r possibly different sets of seeds

[ptions](#) and `Replicates > 1`, then each worker selects seeds and

ans++: The Advantages of Careful Seeding." *SODA '07: Proceedings of
 on Discrete Algorithms*. 2007, pp. 1027–1035.

in PCM." *IEEE Transactions on Information Theory*. Vol. 28, 1982, pp.

broken, NJ: John Wiley & Sons, Inc., 1984.

theory, *FORTTRAN Programs, Examples*. Translated by J. Goldschmidt.

